FPGA Image Processing: VGA Display and a 3x3 Median Filter

Verilog RTL designs on a Xilinx FPGA

FPGA laboratory course – B.Sc. Electrical Engineering, Isfahan University of Technology
Author: Kiavash Rabbani Soudabi (student ID 9821663) – term 4012 (Fall 2022) – individual work

Purpose

Build two hardware image-processing designs in Verilog and verify them through simulation and on the lab board: (1) display a 224x224 image stored in FPGA block memory on a VGA monitor, and (2) filter a stream of image pixels in hardware with a 3x3 median filter, which is the standard pre-processing step before edge detection such as Sobel.

Design 1 – VGA image display

The top level instantiates a clock-wizard IP core that derives the 25 MHz and 40 MHz pixel clocks from the 24 MHz board oscillator, an image ROM initialised from a Xilinx `.coe` file, and a pixel generator that walks the stored image and drives the RGB outputs. Dedicated synchronisation modules generate the horizontal and vertical timing, and a UCF file constrains the pins and the board clock. Test patterns of 224x224 pixels (3-bit and 8-bit) are kept in the repository as ROM initialisation data.

Design 2 – 3x3 median filter

The second design streams pixels from an 8-bit block-memory ROM (depth 50176 = 224x224) and computes the median of every 3x3 neighbourhood on the fly:

- nine registers hold the current window; the column registers are shifted one position per clock after every third clock, so a complete window is always available without a frame buffer;
- the median is extracted with a compare-and-swap network over those registers, and the filtered pixel is written to the output stream together with an `op_finished` flag;
- a 16-bit address counter with a comparison constant marks the end of each three-line block, which keeps the window aligned with the image rows.

Verification

The filter was verified with Verilog testbenches before programming the device; the captured simulation waveform is included in the repository. The ROM images (a noisy photograph used as the median-filter input and a grayscale scene used for edge detection) were decoded from the committed `.coe` files and are shown in the README, so the processed content can be inspected without hardware.

Tools

Verilog-2001, Xilinx ISE (synthesis, UCF constraints, implementation) with ISim/ModelSim for simulation, Xilinx IP cores (clock wizard, block memory generator), VGA display.

Repository contents

`exp6-vga/` (VGA design, pre-lab code, UCF and IP sources) · `exp7-median-filter/` (median filter, testbenches, ROM initialisation) · `data/` (the 224x224 test images in `.coe` format) · `reports/` (submitted pre-report) · `docs/` (this overview, simulation waveform and decoded test images).

Skills demonstrated

Verilog RTL design; FPGA design flow in Xilinx ISE; VGA timing and synchronisation; clock generation with IP cores; block-memory based image storage; streaming image processing in hardware; 3x3 windowing by column shifting; median filter implementation as a compare-and-swap network; testbench-based verification of RTL.